

Day 2: Global Infrastructure + Shared Responsibility

Sunday, 11 October 2026 · 7:00-9:00 PM IST · CLF-C02 tasks 2.1 · 2.2 · 3.1 · 3.2 · Trainer: Gangadhar Ure

Week 1 - Day 2: Global Infrastructure + Shared Responsibility

Live class: Sunday 11 Oct 2026, 7:00-9:00 PM IST Exam focus: Domain 2 (Security and Compliance), Domain 3 (Technology) Exam tasks: 2.1 · 2.2 · 3.1 · 3.2

Before class: make sure root MFA and your budget alert from [Day 1](#) are done. If you got stuck, bring a screenshot and we will help.

Practice Sequence

Seq	Practice	File
05	Regions, AZs, edge locations, ways to access AWS, CloudShell, Health Dashboard	05-global-infrastructure.md
06	Shared responsibility model and compliance (AWS Artifact)	06-shared-responsibility.md

PDF guide: [Day 2 Learner Guide](#)

By The End Of Today

- You can explain Regions, Availability Zones and edge locations, how to choose a Region, and when to use more than one.
- You know the four ways to access AWS and ran your first commands in CloudShell.
- You know what AWS secures, what you secure, and what is shared.
- You know where to find AWS compliance reports (AWS Artifact).

Today's Deliverables

- Screenshot of the Region list and Mumbai AZ table (account ID hidden)
- Four-line Region choice for the fest-registration site
- One line on what the Health Dashboard showed for Mumbai
- One AWS Artifact report name and who might ask for it
- Your 12 shared responsibility card-sort answers in `notes.md`
- Day 2 LinkedIn post: see [10-linkedin-post.md](#)

Self-Check: Exam-Style Questions

Answer first, then open the answer.

1. A company must keep customer data inside India. What should it do first?

A. Use CloudFront · B. Choose an AWS Region in India · C. Use multiple AZs · D. Use AWS Artifact

▼ Answer

B. Data residency and compliance decide the Region.

2. How do you make an application highly available within one Region?

A. Deploy across multiple Availability Zones · B. Use one large EC2 instance · C. Use edge locations only · D. Use AWS Budgets

▼ Answer

A. AZs do not share a single point of failure.

3. Which option lets a team create the same setup again and again in many accounts?

A. AWS Management Console · B. Infrastructure as Code (AWS CloudFormation) · C. AWS Health Dashboard · D. AWS Artifact

▼ Answer

B. Repeatable deployments = IaC.

4. Under the shared responsibility model, who patches the guest operating system on an EC2 instance?

A. AWS · B. The customer · C. Both equally · D. The AWS Partner Network

▼ Answer

B. The customer.

5. Which is a SHARED control?

A. Physical security of data centres · B. Patch management · C. Deciding which Region stores your data · D. Replacing failed disks

▼ Answer

B. Patch management: AWS patches the infrastructure, you patch your guest OS and apps.

6. An auditor asks for AWS ISO and SOC reports. Where do you get them?

A. AWS Trusted Advisor · B. AWS Artifact · C. AWS CloudTrail · D. AWS Health Dashboard

▼ Answer

B. AWS Artifact.

After Day 2

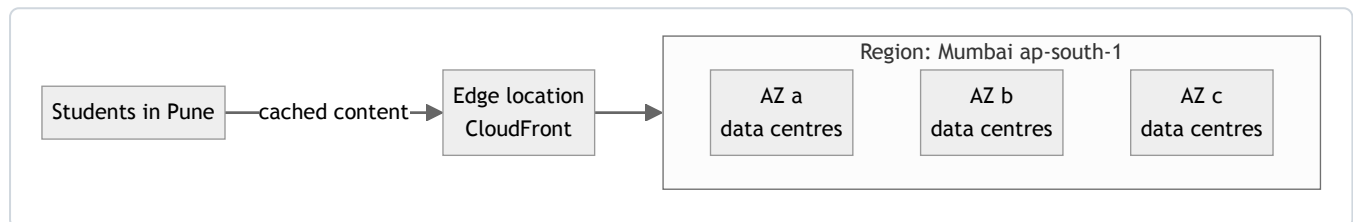
1. Optional: [07-optional-challenge.md](#)
2. Clean up: [08-cleanup.md](#)
3. Submit: [09-submission-format.md](#)
4. Revise: [11-quiz-prep.md](#)

Day 2 - Global Infrastructure and Ways To Access AWS

Exam tasks: 3.1 (ways to access and operate AWS), 3.2 (global infrastructure)

Terms

- **Region:** geographic area with multiple isolated locations (Mumbai `ap-south-1`).
- **Availability Zone (AZ):** one or more data centres in a Region with separate power and networking.
- **Edge location:** site close to users that caches content (CloudFront).
- **Local Zone / Wavelength Zone:** compute closer to a city / inside 5G networks.



Availability Zones are physically separate, with their own power, cooling and networking, connected by fast private links. They **do not share a single point of failure**, so running in two or more AZs keeps you up if one AZ fails.

Live counts are on aws.amazon.com/about-aws/global-infrastructure. Do not memorise numbers; check the live page.

How To Choose A Region

1. Compliance and data residency
2. Latency to users
3. Service availability
4. Cost

When To Use More Than One Region

- Disaster recovery and business continuity (a whole Region has a problem)
- Low latency for users on different continents
- Data sovereignty (each country's data stays in that country)

Exam pointer: data must stay in a country = choose a Region there. Users far away = CloudFront. High availability = multiple AZs. Region-wide disaster recovery = multiple Regions.

Four Ways To Access AWS

Way	What it is	Example
AWS Management Console	Web browser, click to use	What you used on Day 1
AWS CLI	Commands in a terminal	<code>aws ec2 describe-regions</code> in CloudShell

Way	What it is	Example
AWS SDKs	Call AWS from your code (Python, Java, JavaScript...)	A Python app uploading to S3 with <code>boto3</code>
Infrastructure as Code	Describe resources in a file, AWS builds them	AWS CloudFormation, AWS CDK

All four call the same AWS APIs underneath.

Exam pointer: repeatable, automated deployments = Infrastructure as Code (CloudFormation). Access from application code = SDK.

One-time or repeatable?

- One-time task, such as exploring or checking a setting: the Console is fine.
- Same thing many times, or in many accounts or Regions: use the CLI, scripts or IaC.

Practice Rule

Use the Mumbai Region (`ap-south-1`). These labs only **read** information. They create nothing and cost nothing.

Practice in this order:

1. Open CloudShell.
2. Check the AWS CLI.
3. List Regions.
4. List Availability Zones.
5. Check the Health Dashboard.

Lab 1 - First Commands In CloudShell

CloudShell is a terminal in your browser. The AWS CLI is already installed and you are already signed in. It has no extra cost.

1. Sign in to the [AWS Console](#). Check the Region at the top right is **Mumbai**.
2. Click the **CloudShell** icon (a small terminal `>_`) at the top of the Console, next to the search bar. Or search `CloudShell`.
3. Wait for the terminal to load. The first time can take a minute.
4. Type each command and press Enter:

```
aws --version
aws configure list
```

Expected result:

- `aws --version` shows a version number, for example `aws-cli/2.x.x`.
- `aws configure list` shows where your credentials come from. You did not type any key.

Do not post `aws sts get-caller-identity` output; it shows your account ID.

Lab 2 - Region And AZ Explorer

Run each command one by one in CloudShell:

```
aws ec2 describe-regions --query "Regions[].RegionName" --output text
```

Expected result: a list of Region codes. Find `ap-south-1` (Mumbai).

```
aws ec2 describe-regions --all-regions --query "Regions[][RegionName,OptInStatus]" --output table
```

Expected result: a table. Some Regions show `not-opted-in`, for example Hyderabad `ap-south-2`. Do **not** enable them for this lab.

```
aws ec2 describe-availability-zones --region ap-south-1 --query "AvailabilityZones[][ZoneName,ZoneId,State]" --output table
```

Expected result: Mumbai AZs such as `ap-south-1a`, each with a zone ID such as `aps1-az1` and state `available`. Take your screenshot here.

```
aws ec2 describe-availability-zones --region us-east-1 --query "AvailabilityZones[].ZoneName" --output text
```

Expected result: N. Virginia AZ names. Different Region, different AZs.

Notes:

- Opt-in Regions (such as Hyderabad `ap-south-2`) must be enabled before use. Do not enable them for this lab.
- Zone names are mapped per account; zone IDs are the same for everyone.

Close CloudShell when you finish.

Lab 3 - AWS Health Dashboard

1. Search `Health` in the Console and open **AWS Health Dashboard**.
2. Open **Service health**. Filter by **Asia Pacific (Mumbai)**.
3. Open **Your account health**. Check for open issues or scheduled changes.

Expected result: you can write one line about what the dashboard shows for Mumbai. Often it shows no current issues.

Health Dashboard shows AWS events that may affect you. It costs nothing.

Lab 4 - Choose A Region For The Fest Site

Scenario: a fest-registration site for students in Pune. Student data must stay in India.

Write four lines in your `notes.md`, one for each factor:

Compliance:
Latency:
Service availability:
Cost:
My Region choice:

Tip: check the [AWS global infrastructure map](#) and the [Regional services list](#).

Deliverables

- Screenshot of the Region list output
- Screenshot of the Mumbai AZ table (account ID hidden)
- Four-line Region choice for a fest-registration site for students in Pune
- One line: what did the Health Dashboard show for Mumbai?

If You Get Stuck

Problem	Try this
CloudShell keeps loading	Wait 2 minutes, refresh the page, check the Region is Mumbai
UnauthorizedOperation error	Sign in as the root user for this lab (IAM users come in Week 2)
My <code>ap-south-1a</code> is different from my friend's	Normal. Compare the Zoneld column instead

I listed the Regions. CloudShell would not load for the AZ command, so I will retry tomorrow.

Official Links

Topic	Link
AWS global infrastructure	https://aws.amazon.com/about-aws/global-infrastructure/
Regions and Availability Zones	https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-regions-availability-zones.html
Services available in each Region	https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services/
Availability Zone IDs	https://docs.aws.amazon.com/ram/latest/userguide/working-with-az-ids.html
Amazon CloudFront (edge locations)	https://aws.amazon.com/cloudfront/
AWS CloudShell	https://docs.aws.amazon.com/cloudshell/latest/userguide/welcome.html
AWS CLI	https://aws.amazon.com/cli/
AWS SDKs and tools	https://aws.amazon.com/developer/tools/
AWS CloudFormation (Infrastructure as Code)	https://aws.amazon.com/cloudformation/
AWS Health	https://docs.aws.amazon.com/health/latest/ug/what-is-aws-health.html

Day 2 - Shared Responsibility Model

Exam tasks: 2.1 (shared responsibility), 2.2 (compliance and governance)

AWS secures the cloud. You secure what you build in the cloud.

	AWS	You
Headline	Security of the cloud	Security in the cloud
Examples	Data centres, hardware, hypervisor, global network	Data, IAM, firewall rules, EC2 guest OS patching

Responsibility Shifts By Service

Service	AWS handles	You handle
EC2	Hardware, virtualisation	Guest OS, patches, apps, security groups, data
RDS	Hardware, OS, database patching	Data, access, network access, settings
Lambda	Servers, OS, runtime	Code, permissions, data
S3	Hardware, service durability	Bucket permissions, encryption choice, content

Controls: Inherited, Shared, Customer

Type	Meaning	Examples
Inherited	AWS does it fully; you inherit it	Physical and environmental security of data centres
Shared	Both do their part, each in its own layer	Patch management (AWS patches infrastructure, you patch your guest OS and apps), configuration management , awareness and training
Customer-specific	Only you	Your data, who has access, which Region stores it

Exam pointer: the more managed the service (EC2 -> RDS -> Lambda), the less you manage.

Compliance: Where To Find The Proof

- **AWS Artifact**: free, self-service portal for AWS compliance reports (such as ISO and SOC) and agreements.
- **AWS Compliance Programs** (aws.amazon.com/compliance): which standards AWS meets by industry and country.
- AWS being compliant does **not** make your app compliant. You still configure your part.

Lab - Find A Compliance Report In AWS Artifact

Practice in this order:

1. Read the [shared responsibility model](#) page. Find one thing AWS does and one thing you do.

2. Sign in to the [AWS Console](#). Region: Mumbai.
3. Search **Artifact** and open **AWS Artifact**.
4. Click **Reports** in the left menu.
5. In the search box, type **ISO** . Then try **SOC** .
6. Click one report name. Read its title and description.
7. Do **not** download or accept any agreement. Reading is enough.

Expected result: you can name one report and say who might ask for it.

Deliverable: one line naming a report you found and who might ask for it (for example, a college or bank client).

If you get stuck:

I opened AWS Artifact but the Reports page did not load. I read the shared responsibility page instead.

Activity - Card Sort

Write A (AWS) or C (Customer) for each. Try first, then open the answers below.

1. Replacing a failed disk in a data centre
2. Deciding who can read your S3 bucket
3. Physical security of the building
4. Patching the OS on your EC2 instance
5. Patching the database engine on RDS
6. Encrypting your customer data
7. Keeping the global network cables running
8. Rotating your IAM user passwords
9. Configuring your security group rules
10. Keeping the Lambda runtime up to date
11. Deciding which Region stores your data
12. Hypervisor security

▼ Show answers

#	Answer	Why
1	A	Physical hardware
2	C	Access to your data
3	A	Physical security
4	C	Guest OS on EC2 is yours
5	A	RDS is managed; AWS patches the engine
6	C	You choose to encrypt your data
7	A	Global network
8	C	Your IAM users and passwords
9	C	Your firewall rules

#	Answer	Why
10	A	Lambda is serverless; AWS manages the runtime
11	C	You choose the Region
12	A	Virtualisation layer

Deliverable: your 12 answers in `notes.md` , with one line on any you got wrong and why.

Official Links

Topic	Link
Shared responsibility model	https://aws.amazon.com/compliance/shared-responsibility-model/
AWS Artifact	https://docs.aws.amazon.com/artifact/latest/ug/what-is-aws-artifact.html
AWS compliance programs	https://aws.amazon.com/compliance/programs/
AWS Cloud security	https://aws.amazon.com/security/

github.com/cloudadhar/cloud-unlock-aws-challenge · youtube.com/@CloudAdhar · Use synthetic data only. Never share account IDs, access keys, MFA QR codes, OTPs or card details.